PdfHelperReader – Comprehensive User Manual

(v3.1.0) • High-Performance Dual-Pane Reading, LLM Vocabulary Engine & Integrated Research Environment

1. Welcome & Introduction

PdfHelperReader is a specialized Android document reading environment designed for language learning, academic research, and deep study. It reads both PDF and EPUB documents, pairing your main book with an LLM-refined companion vocabulary sidecar PDF or an embedded Google Web Search browser in real-time.

2. Upper Main Ribbon Controls (Top Action Bar)

Button	Function & Detailed How-To-Use Guide
B	**Add Bookmark**: Tap to bookmark your current reading position. Captures creation timestamp and extracts the first 25 words from the book page.
BL	**Bookmark List**: Opens saved bookmarks dialog. Shows creation date/time and 25-word book page snippet previews for the current book.
Both / Book / Vocab	**Split Mode Toggle**: Cycles screen layout between Both (dual-pane split view), Book (full-screen main document), or Vocab (full-screen vocabulary pane).
Z	**Zen Mode**: Hides all toolbars for distraction-free reading, replacing them with a docked vertical right-edge control strip.
SR	**Synced Reading**: Toggles 1:1 real-time bidirectional scroll synchronization. When scrolling either the Book or Vocab pane, the other pane scrolls automatically in lockstep.
AS	**Auto Scroll**: Gentle continuous frame-by-frame auto-scroll calibrated at 1 viewport height / 5 minutes. Touch dragging is supported without stopping auto-scroll.
Sync	**Page Sync**: Instantly jumps the Vocabulary pane to match your current Book page using the exact vocab_map array.
G:Vocab / G:Google	**Vocab Mode**: Toggles lower pane between Vocabulary PDF sidecar and embedded Google Search. Automatically switches to Dual-Pane Split View if in Single Document Mode.
T	**Theme Toggle**: Switches Dark/Light mode with high-contrast ColorMatrix PDF page color inversion for comfortable night reading.
R:C / R:P	**Reading Mode**: Toggles Continuous vertical scrolling (R:C) or fit-to-width Page-by-Page mode (R:P).
V:PDF / V:TXT	**View Mode**: Toggles vector PDF page rendering with transparent OCR text selection overlay (V:PDF) or plain text extraction (V:TXT).
H	**History**: Opens reading history showing First Opened & Last Closed timestamps for up to 1,000 documents, with a Del vocab button to delete associated vocabulary files.
O	**OCR Generator**: Applies Google ML Kit OCR to image-based or scanned PDFs, adding a searchable text layer saved to Downloads/PdfHelperReader_files/.
LOB	**Load Online Book**: Opens dual-tab online library (35 GitHub CDN classics + official Project Gutenberg search engine for 70,000+ books with EPUB and PDF download options).
OB / OV / LV	**Open Book / Open Vocab / Load Vocab**: Open PDF or EPUB book/vocab files via system file picker, or generate vocabulary PDF automatically.
Update	**In-App Update**: Checks GitHub latest release API, streams live download progress, and launches Android Package Installer seamlessly.
App Info	**Developer Info**: Displays developer profile (Ohid Sajol), GitHub repository links, and application overview.

3. LLM-Refined Vocabulary Engine & Difficulty Scores

PdfHelperReader features a fine-tuned dictionary database containing over 800,000 entries. Generated vocabulary PDFs format each word with parts of speech (noun, verb, adj), definitions, example usage, synonyms, antonyms, and difficulty scores displayed right beside headwords in slate gray (e.g. **anopheles (29.25)**). Automatic streaming initialization ensures zero OOM crashes even on large documents.

4. Alternating #98E8DE Styling & Auto Split Switching

Main ribbon buttons alternate dynamically between primary container and vibrant #98E8DE (Soft Teal) Material 3 container themes. Activating Google Search mode automatically switches Single Document Mode to Dual-Pane Split View.

5. Instant Copy-to-Search Popup Banner

Whenever you select and copy text anywhere in the app or system, a floating popup banner appears allowing you to search the copied string in Google with a single tap. The banner responds reliably to every copy action, even when re-copying the same selection.

6. In-App Updates & Developer Info

Tapping Update checks GitHub for new release versions, downloads the APK in the background with a live progress dialog, and launches the Package Installer. Tapping App Info displays developer details and repository links.

7. Gutenberg Books Search Engine (LOB Tab 2)

Tapping LOB opens the dual-tab Online Library. The Gutenberg Books tab features live search across 70,000+ books with direct [EPUB] and [PDF] download options.

8. Full EPUB eBook Support

PdfHelperReader unzips and converts eBooks on-demand into clean, typeset vector documents with large 18pt typography. Converted files are cached permanently in Downloads/PdfHelperReader_files/ so re-opening an EPUB loads instantly in 0ms.

9. 500 MB Storage Manager

An automatic 500 MB Storage Manager monitors folder size, deleting older files step-by-step while protecting the latest active file in each category.